Trigonometry: Inverse Trigonometric Functions

Questions 1–27

Question 1. Sketch the following graphs, in radians, identifying the coordinates of any endpoints and the equations of any asymptotes.

a) $y = \arcsin x$

b) $y = \arccos x$

c) $y = \arctan x$

Question 2. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = 3\arcsin x$

b) $y = \frac{1}{4}\arcsin x$

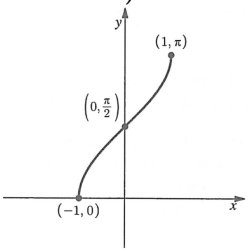
Question 3. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \arcsin 3x$

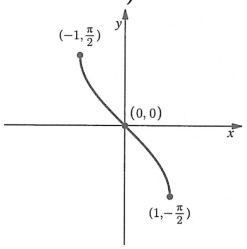
b) $y = \arcsin \frac{2}{5}x$

Question 4. The following are all functions that are single transformations of $y = \arcsin x$. Write down the equation in each case.

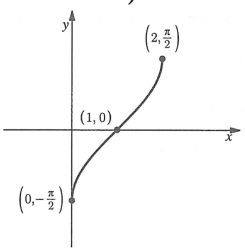
a)



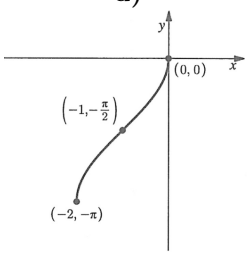
b)



c)



d)



Question 5. Sketch the graph of $y = -\arcsin(x+1)$ in radians, identifying the coordinates of any endpoints.

Question 6. Sketch the graph of $y = \frac{\pi}{2} - \arcsin x$ in radians, identifying the coordinates of any endpoints.

Question 7. It is given that $f(x) = \arcsin x$, where $-1 \leq x \leq 1$. Find the exact value for which $\pi - 3f(2x-3) = 0$.

Question 8. Prerequisite knowledge for Q8: Domain and Range and Inverse Functions It is given that $f(x) = 2 - 3\sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

a) Show that $f^{-1}(x) = \arcsin\left(\frac{2-x}{3}\right)$.

b) State the domain and range of $y = f^{-1}(x)$.

Question 9. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \frac{5}{2}\arccos x$

b) $y = \frac{1}{5}\arccos x$

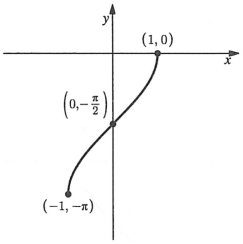
Question 10. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \arccos 4x$

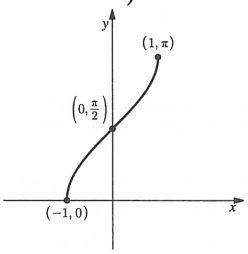
b) $y = \arccos \frac{4}{5}x$

Question 11. The following are all functions that are single transformations of $y = \arccos x$. Write down the equation in each case.

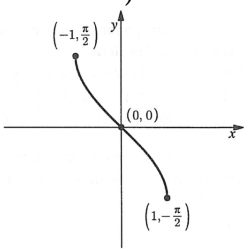
a)



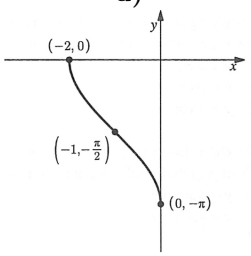
b)



c)



d)



Question 12. Sketch the graph of $y = -\arccos(x - 1)$ in radians, identifying the coordinates of any endpoints.

Question 13. Sketch the graph of $y = \arccos 2x - \pi$ in radians, identifying the coordinates of any endpoints.

Question 14. It is given that $g(x) = \arccos x$, where $-1 \leq x \leq 1$. Find the exact value for which $6\pi - 9g(0.2x + 3) = 0$.

Question 15.

a) Solve the equation $\arccos x = \frac{\pi}{3}$.

b) Hence find the exact value of $\arcsin x$.

Question 16. Sketch the following graphs, in radians, identifying the equations of any asymptotes:

a) $y = 8 \arctan x$

b) $y = \frac{1}{6} \arctan x$

Question 17. Sketch the following graphs, in radians, identifying the equations of any asymptotes:

a) $y = 2 + 3\arctan x$

b) $y = 5 - 2\arctan x$

Question 18. Sketch the following graphs, in radians, identifying any endpoints and the equations of any asymptotes:

a) $y = 3\arcsin x + \pi$

b) $y = \arccos(\frac{1}{2}x - 2)$

c) $y = 2\arctan x - \frac{\pi}{3}$

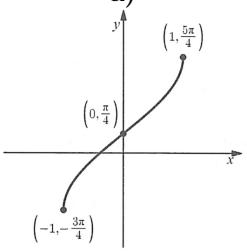
d) $y = \arcsin(4x + 5)$

e) $y = \frac{2\pi}{3} - \arccos x$

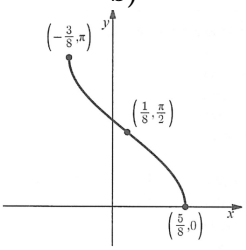
f) $y = 3\pi - \arctan x$

Question 19. Write down **two** suitable equations for the following graphs:

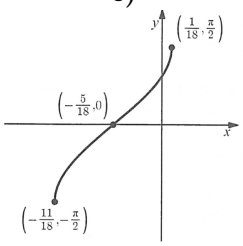
a)



b)



c)



Question 20. Solve the equation $\arcsin x = \arccos x$.

Question 21. Prerequisite knowledge for Q21: Double Angle Formulae

a) Solve the equation $\arcsin x = 2\arccos x$.

b) Solve the equation $2\arcsin x = \arccos x$.

Question 22. Given $\arcsin x = \frac{\pi}{5}$, show that $x = \sqrt{\frac{5-\sqrt{5}}{8}}$.

The remaining questions in this exercise are all in radians and are to be completed without a calculator.

Question 23. Find

a) $\arcsin 1$

b) $\arccos(-1)$

c) $\arcsin 0$

d) $\arccos 1$

Question 24. Find

a) $\arcsin \frac{\sqrt{2}}{2}$

b) $\arccos \frac{\sqrt{2}}{2}$

c) $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

d) $\arccos\left(-\frac{\sqrt{2}}{2}\right)$

Question 25. Find

a) $\arctan 1$

b) $\arctan(-1)$

c) $\arctan \sqrt{3}$

d) $\arctan\left(-\frac{\sqrt{3}}{3}\right)$

Question 26. Find

a) $\arcsin\left(\sin \frac{\pi}{2}\right)$

b) $\arccos\left(\cos\left(-\frac{\pi}{2}\right)\right)$

c) $\arcsin\left(\sin \frac{3\pi}{2}\right)$

d) $\arccos\left(\cos\left(-\frac{3\pi}{2}\right)\right)$

e) $\arcsin\left(\sin \frac{\pi}{3}\right)$

f) $\arccos\left(\cos \frac{2\pi}{3}\right)$

g) $\arcsin\left(\sin \frac{5\pi}{3}\right)$

h) $\arccos\left(\cos\left(-\frac{\pi}{3}\right)\right)$

i) $\arctan\left(\tan \frac{\pi}{4}\right)$

j) $\arctan\left(\tan \frac{2\pi}{3}\right)$

Question 27. Find

a) $\sin\left(\arcsin\frac{\sqrt{2}}{2}\right)$

b) $\cos\left(\arccos\left(-\frac{\sqrt{2}}{2}\right)\right)$

c) $\sin\left(\arccos\frac{1}{2}\right)$

d) $\cos\left(\arcsin\frac{1}{2}\right)$

e) $\sin\left(\arccos\frac{\sqrt{3}}{2}\right)$

f) $\cos\left(\arcsin\left(-\frac{\sqrt{3}}{2}\right)\right)$

g) $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$